

CITS5206 Information Technology Capstone Project
Deliverable 3 – Group 14, 2026

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1. Summary of Project Requirements and Proposed Solution(s)

1.1 Project Requirements

The client, Peter Whittaker from the Mechanical Engineering department at UWA with industry partner Programmed, needed a system that could give pump operators an early warning before a fault develops — not a system that detects faults after they have already happened. The key distinction confirmed with the client was prediction, not detection: the system had to look ahead using recent sensor history and flag what was coming, not describe what was currently happening.

Requirements were gathered across all client meetings and formally documented in a Software Requirements Specification. They are divided into functional requirements (what the system must do) and non-functional requirements (how it must behave).

Functional Requirements — Data Input

ID	Requirement	Priority
FR-01	The system shall provide a web page on which the user can upload a CSV file by either drag-and-drop or click-to-browse.	Must
FR-02	The system shall validate the uploaded file against the SKAB column schema before any model is invoked.	Must
FR-03	If the uploaded file does not conform to the expected schema, the system shall display a plain-language error message identifying which columns are missing, extra, or out of order. It shall not attempt automatic correction.	Must
FR-04	When a dataset has fewer sensor channels than the SKAB training set, the system shall reject the file with an explanatory error message rather than padding missing channels with synthetic values.	Must
FR-05	The system shall provide a downloadable sample CSV file showing the expected schema, accessible from the upload page.	Must
FR-06	The site navigation shall use a tab layout (Home, Upload, Results) replacing the current top-right Get Started button.	Should
FR-07	On a successful upload, the system shall display a confirmation showing the file name, number of rows accepted, and the time range covered.	Should

Functional Requirements — Model Selection and Prediction

ID	Requirement	Priority
FR-08	The system shall offer the user a choice of all eight pre-trained models, each presented with its name and headline performance metrics (Recall, Precision, F1) on the SKAB held-out test set.	Must
FR-09	The system shall apply the same StandardScaler fitted on the training split when transforming uploaded data, so inference-time and training-time inputs are on the same scale.	Must
FR-10	The system shall construct sliding windows over the uploaded data using window length 20 steps and step size 1, matching the training-time configuration.	Must

ID	Requirement	Priority
FR-11	On clicking Run Prediction, the system shall invoke the selected model and return one anomaly probability per window. A loading indicator shall be shown while prediction is running.	Must
FR-12	The system shall complete prediction on a 1-hour input file within 30 seconds on the development environment.	Should
FR-13	The system should allow the user to select an alternative prediction horizon between 10 seconds and 10 minutes, with a notice that metrics are based on the trained 10-step horizon.	Could

Functional Requirements — Results Presentation

ID	Requirement	Priority
FR-14	The system shall display anomaly probability over time as a line chart. The X-axis represents time and the Y-axis represents probability in the range 0 to 1.	Must
FR-15	Time windows whose probability exceeds the alert threshold shall be visually highlighted so that anomalous regions are identifiable at a glance.	Must
FR-16	The default alert threshold shall be 0.5. The threshold shall be a configuration value rather than a hard-coded constant.	Must
FR-17	The system shall offer a PDF report download covering the chart, the model used, headline metrics, the anomaly summary panel, and the input file name.	Must
FR-18	The PDF report shall additionally summarise the input columns detected, preprocessing steps applied, and a plain-language explanation of the chart for non-technical readers.	Should
FR-19	Below the main chart, the system shall display a side-by-side comparison view showing the anomaly curves of all eight models on the same input.	Should
FR-20	After prediction completes, the system shall show a summary panel reporting: total number of windows, number of anomaly windows, anomaly time ranges, and the time of peak probability.	Should
FR-21	The system shall provide an unlabelled-data mode. When the input file has no anomaly column, the system shall run prediction normally and suppress metrics that require ground truth.	Should

Non-Functional Requirements

The non-functional requirements set the standards for how the system behaves across seven dimensions: usability, reliability, performance, maintainability, portability, security, and accessibility.

ID	Category	Requirement
NFR-01 (Must)	Usability	The system shall be usable by a person with general technical literacy and no prior machine learning knowledge. Domain-specific terms shall be avoided or explained with tooltips.

ID	Category	Requirement
NFR-02 (Should)	Usability	All interactive elements shall provide a hover state and a tooltip describing their function.
NFR-03 (Should)	Usability	The system shall provide a dark/light theme toggle, with the user's preference persisted across sessions.
NFR-04 (Must)	Reliability	The system shall not crash on malformed input. All foreseeable input errors shall produce user-facing error messages.
NFR-05 (Must)	Reliability	Inference results shall be reproducible. The same model on the same input shall return identical predictions.
NFR-06 (Should)	Performance	The landing page shall load in under 2 seconds on a standard broadband connection.
NFR-07 (Should)	Performance	Prediction on a 1-hour SKAB-format file shall complete within 30 seconds for any of the eight bundled models.
NFR-08 (Must)	Maintainability	Trained model files shall be stored as versioned artefacts. The expected schema shall be documented alongside them.
NFR-09 (Should)	Maintainability	The codebase shall separate frontend, API layer, and model layer into distinct modules so any one can be replaced without touching the others.
NFR-10 (Should)	Portability	The system shall be packaged as a Docker container. A docker-compose configuration shall bring the system up in a single command.
NFR-11 (Could)	Portability	The system should run unchanged on Linux, macOS, and Windows via Docker.
NFR-12 (Must)	Security	The system shall not transmit uploaded data to any third-party service. All processing shall be local to the deployment environment.
NFR-13 (Should)	Security	The system shall not retain uploaded files beyond the current user session unless the user explicitly downloads the report.
NFR-14 (Could)	Accessibility	Colour choices in charts and badges shall be distinguishable for users with common forms of colour vision deficiency, and shall not rely on colour alone to convey state.